

GERALDTON, Australia

WMO: 944030

Lat: 28.8044S

Lon: 114.6989E

Elev: 30

StdP: 100.96

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	6.5	-6.1	2.3	24.4	-3.2	2.9	23.7	12.2	17.3	11.1	17.4	3.2	130	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.8	38.3	19.5	36.1	19.2	33.9	19.0	23.5	29.8	22.7	28.6	22.0	27.6	7.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	16.7	25.5	21.1	15.8	25.0	20.2	15.0	24.5	70.6	29.7	67.3	28.7	64.7	27.6	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	11.2	10.3	3.0	42.6	1.0	1.5	2.2	43.7	1.6	44.6	1.1	45.4	0.3	46.5		
(5)			2.2	25.3	1.0	2.1	1.5	26.8	0.9	28.0	0.3	29.2	-0.4	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	24.6	25.6	24.5	21.7	18.8	16.1	14.5	14.6	15.5	17.9	20.9	23.0		
	DBStd	4.80	3.04	3.46	3.35	2.79	2.45	2.13	2.10	2.09	2.26	2.67	3.01	3.27		
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0		
	HDD18.3	471	0	0	0	3	23	72	119	118	91	39	6	1		
	CDD10.0	3569	452	437	451	351	273	182	141	142	164	246	326	404		
	CDD18.3	998	193	204	193	104	37	4	1	2	5	27	82	147		
	CDH23.3	9066	1737	2032	1766	772	249	29	6	20	82	291	749	1333		
(13)	CDH26.7	3984	798	994	806	272	53	2	0	3	20	102	323	613		
(14)	Wind	WSAvg	5.5	6.8	6.5	6.0	5.0	4.7	4.6	4.5	4.7	5.4	6.2	6.8		
(15) Precipitation	PrecAvg	436	5	8	16	24	67	93	90	64	33	18	10	5		
	PrecMax	755	64	84	134	75	282	256	185	155	76	101	47	23		
	PrecMin	196	0	0	0	0	9	6	23	8	4	0	0	0		
	PrecStd	111	11	17	25	21	50	54	38	25	17	17	12	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	40.9	39.1	35.0	30.8	26.1	24.0	26.0	29.3	33.8	37.8	40.1		
		MCWB	20.2	20.0	19.7	17.3	15.8	12.9	14.2	14.1	15.1	16.3	17.4	19.3		
	2%	DB	37.2	37.8	36.2	32.2	28.0	23.9	21.8	22.8	25.6	29.4	33.9	36.6		
		MCWB	19.8	20.2	19.3	17.1	15.4	13.7	14.4	15.1	14.7	15.6	16.5	18.6		
	5%	DB	33.9	35.3	33.7	29.8	26.0	22.2	20.3	21.1	23.0	26.2	30.3	32.9		
		MCWB	19.9	20.3	19.0	17.6	15.4	14.4	14.5	15.1	15.4	15.8	16.7	18.4		
	10%	DB	30.9	32.7	31.1	27.6	24.2	20.8	19.2	19.8	21.2	23.9	27.2	29.8		
		MCWB	19.9	20.4	19.1	17.9	15.8	14.3	14.2	14.6	15.0	16.1	17.3	18.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	25.0	23.7	22.2	20.7	18.6	18.7	18.6	18.6	19.6	21.0	22.9		
		MCDB	30.2	31.4	28.5	25.8	23.1	20.9	20.3	21.3	22.5	24.8	26.3	30.2		
	2%	WB	23.2	23.9	22.7	21.3	19.6	17.6	17.0	17.3	17.5	18.6	20.0	21.6		
		MCDB	29.4	30.6	28.1	24.9	22.6	20.1	19.1	20.1	21.5	23.5	25.4	28.0		
	5%	WB	22.3	23.0	22.1	20.6	18.5	16.6	15.9	16.2	16.6	17.9	19.2	20.9		
		MCDB	28.3	29.1	27.3	24.5	22.2	19.5	18.4	19.4	20.7	22.8	25.0	27.3		
	10%	WB	21.4	22.2	21.4	19.8	17.6	15.7	15.1	15.3	15.8	17.2	18.6	20.2		
		MCDB	27.3	27.9	26.6	24.1	21.7	18.8	17.9	18.5	19.8	22.0	24.3	26.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.8	11.9	11.8	11.1	10.0	9.9	11.0	12.4	12.8	12.2	11.8		
		MCDBR	17.8	16.2	15.8	15.1	13.7	12.2	11.8	13.3	16.1	17.8	18.7	18.6		
		MCWBR	5.7	5.2	5.8	6.3	6.4	6.3	6.9	7.7	8.1	7.7	6.5	6.3		
	5% WB	MCDBR	11.4	11.3	10.8	10.7	9.9	8.7	8.9	10.8	12.7	12.9	12.5	12.3		
		MCWBR	4.2	4.2	4.5	5.4	5.8	5.7	6.4	7.1	7.3	6.4	5.4	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.353	0.337	0.323	0.305	0.295	0.295	0.301	0.318	0.324	0.330	0.337		
	taud		2.548	2.574	2.596	2.619	2.637	2.649	2.634	2.612	2.542	2.547	2.549	2.556		
	Ebn at Noon		989	970	956	923	896	884	897	933	959	989	1004	1007		
	Edh at Noon		110	104	96	85	75	71	74	84	99	105	109	109		
(44) All-Sky Solar Radiation	RadAvg		8.15	7.47	6.32	4.90	3.81	3.19	3.40	4.36	5.66	7.02	7.97	8.47		
	RadStd		0.43	0.31	0.38	0.25	0.22	0.20	0.18	0.13	0.25	0.24	0.21	0.30		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page